

Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

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Advanced Algorithms and Parallel Programming – 2nd part (Parallel Programming)–June 9th 2022

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Questions 1 (4 points) _____

Questions 2 (4 points) _____

Questions 3 (4 points) _____

Questions 4 (4 points) _____

Questions n. 1

Answer the following questions about OpenMP and briefly explain (without explanation the answer will be considered invalid)

A. Are there any compiler-related limitations to the use of OpenMP?

B. What are Work-Sharing constructs in OpenMP? Describe their use and the conditions required for them to effectively parallelize a task.

C. Describe how to parallelize the search of an element inside a list of unknown length using OpenMP.

D. Is it possible to express nested parallelism in OpenMP? Is it generally a good practice?

Questions n. 2

Answer the following questions about parallel algorithms and parallel programming and briefly explain (without explanation the answer will be considered invalid)

A. Is it possible to obtain the same low level implementation starting from different parallel programming models descriptions of the same parallel pattern?

B. Parallel programming models using threads spawn a single process, therefore the program I/O correctness is always guaranteed: true or false?

C. Describe the scan parallel pattern. Can it be implemented using different approaches with respect to the computation of intermediate values?

Questions n. 3

Answer the following questions about MPI and briefly explain (without explanation the answer will be considered invalid)

A. Which memory architectures are supported by MPI? Are there any differences with the other parallel programming models presented?

B. Describe the behavior of MPI_Gather.

C. Describe how to spread data across processes according to a user-defined distribution using MPI.

Questions n. 4

Answer the following questions about PThread and briefly explain (without explanation the answer will be considered invalid)

A. Does PThread use an explicit or implicit synchronization model? Describe common approaches to manage the access to critical sections.

B. What are the possible approaches for managing the execution flow of different threads using PThreads?